DIFFERENTIAL - TRU - LOK

DESCRIPTION

The differential is a locking differential, that provides a positive mechanical connection between the right and left axle when engauged. The differential uses a dog clutch to connection the right and left axle.

OPERATION

The Tru-lok differential is activated by the axle lock switch located on the dash panel. When the switch is activated, an air pump with a built-in pressure regulator sends 5 PSI of air pressure to a accuator diaphragm in the differential housing. The diaphragm then engages a dog clutch and a position switch. The dog clutch has one gear attached to the differential case and another gear attached to a differential side gear. When the dog clutch is engaged the right and left wheels turn at the same speed. The position switch lights a lamp on the dash to indicate the system has been engaged. The differential works as standard differential when not engaged.

NOTE: The differential is serviced as an assembly, the diaphragm and indicator switch are serviced separately. The differential case must be removed to service the diaphragm actuator and indicator switch.

DIAGNOSIS AND TESTING

UNLOCKED

(1) Block tires opposite the axle to be tested to prevent the vehicle from rolling.

(2) Place transfer case in 4WD Low and automantic transmission in Park (1st gear if manual transmission).

(3) Raise both wheels of the axle to be tested off the ground.

(4) Turn ignition to the ON position and dash switch to the OFF position.

(5) Rotate one tire by hand. The other tire should spin in the opposite direction.

NOTE: If wheel cannot be rotated the differential must be repaired/replaced.

LOCKED

(1) Block tires opposite the axle to be tested to prevent the vehicle from rolling.

(2) Place transfer case in 4WD Low and automantic transmission in Park (1st gear if manual transmission).

(3) Raise both wheels of the axle to be tested off the ground.

(4) Turn ignition to the ON position and dash switch to the ON position.

(5) Try to rotate one tire by hand. You should not be able to rotate the tire.

NOTE: If wheel does rotate verify locker pump operation. If the pump is operating properly the differential must be repaired/replaced.

TRU - LOK INDICATOR SWITCH

(1) Turn ignition switch off.

(2) Disconnect indicator switch harness (Fig. 44) from the differential housing.

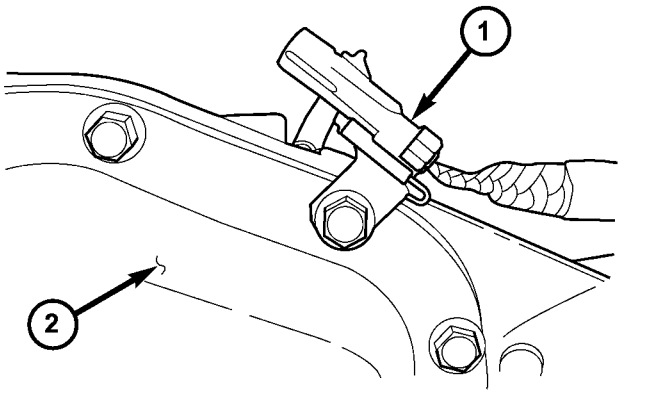

Fig. 44 INDICATOR SWITCH CONNECTOR

1 - CONNECTOR
2 - DIFFERENTIAL COVER

(3) Measure electrical continuity across the switch terminals. Circuit should be closed (zero resistance).

NOTE: If circuit is not closed remove differential and replace locker indicator switch.

TRU - LOK PUMPS

(1) Connect a pressure gauge to the front pump.

(2) Place transfer case in 4WD Low and automatic transmission in Park (1st gear if manual transmission).

(3) Turn ignition to the ON position and push the dash switch twice to activate the front pump.

NOTE: If pump is not running, verify pump has (Fig. 45) 12 volts and a ground.